

Neptune

Neptune is the eighth and most distant planet from the Sun, orbiting at about 30 astronomical units and taking 165 Earth years to complete one orbit. It was the first planet found by mathematical prediction rather than observation: Urbain Le Verrier calculated its position from irregularities in the orbit of Uranus, and Johann Galle observed it in 1846 within a degree of the predicted spot.

Neptune is an ice giant slightly smaller than Uranus, with an equatorial diameter of about 49,528 kilometres, but more massive. Its deep blue colour also comes from methane. Neptune has the strongest sustained winds in the Solar System, measured at around 2,000 kilometres per hour.

When Voyager 2 flew past in August 1989 it photographed a large storm called the Great Dark Spot, comparable in size to Earth. When the Hubble Space Telescope looked in 1994 the spot had vanished, and other dark spots have appeared and disappeared since.

Neptune has 16 known moons. The largest, Triton, orbits in the retrograde direction, opposite to Neptune's rotation, which strongly suggests it is a captured Kuiper belt object rather than a moon that formed in place. Voyager 2 saw active geysers of nitrogen on Triton's surface, making it one of the few geologically active bodies in the outer Solar System.